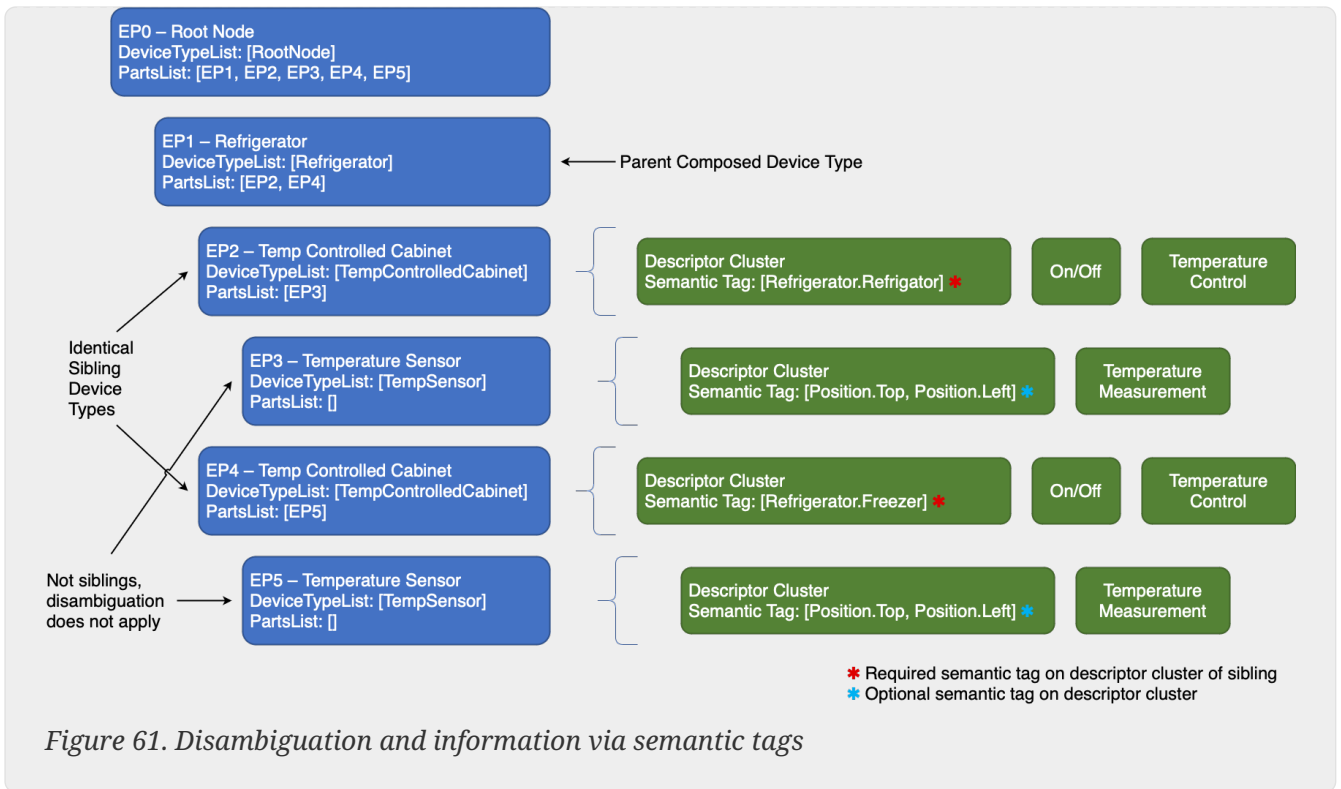




9.2.10. Dynamic Endpoint Allocation

Some nodes MAY require a dynamic number of endpoints, since the functionality they expose can change at run-time, e.g.

- a [Bridge](#) on which Bridged Devices are added or deleted.
- a Casting Video Player in which Content App endpoints are added or deleted (see Device Library, section 10).

Such dynamic entities which need to be exposed with an endpoint, will be referred to as an "exposed entity" in the following description.

For any node which changes its endpoint composition, the endpoint addresses SHALL be allocated according to the following rules:

- When such exposed entity is exposed for the first time, it SHALL be allocated a *new* endpoint address (or set of endpoint addresses), incrementing from the highest previously (ever) used endpoint address.
 - For the situation where a node following these mechanisms has exhausted all available remaining endpoint addresses for exposed entities, it MAY wrap around to the lowest unused endpoint address, starting at 1, since the RootNode endpoint (endpoint 0) cannot be used to host application device types and is therefore never free.
- For existing exposed entities, the endpoint addresses SHALL NOT be changed.
 - This persistency requirement also holds for the case of restart/reboot of the device.

As a result of these mechanisms, endpoint addresses that were used for exposed entities that were once exposed but now have been removed will *not* be reused in the future (apart from the exceptional wrap-around corner case mentioned above), in order to avoid the possibility of confusing